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Abstract

The purpose of the following project is to present a machine learning pipeline for binary classification of coronary artery disease using the UCI Heart Dataset (Cleveland subset), comprising 242 patient records and 13 clinical features. The project begins with an exploratory data analysis, followed by a repeated cross-validation framework using object-oriented programming principles to provide unbiased estimates of generalization performance while enabling systematic hyperparameter optimization via Optuna. Seven classifiers were evaluated: Logistic Regression with Elastic Net regularization, Gaussian Naive Bayes, Linear Discriminant Analysis, Random Forest, LightGBM, XGBoost, and CatBoost.

All preprocessing steps including missing value imputation, categorical encoding, and feature scaling, were applied strictly within each fold to prevent data leakage and stratified splitting was used throughout to preserve class distribution. The classifiers were compared based on clinically motivated metrics including the Matthews Correlation Coefficient (MCC), Area Under the ROC Curve (AUC), Balanced Accuracy, F1 score, Recall, Specificity, and Precision-Recall AUC, with 95% bootstrap confidence intervals computed over 50 outer-fold evaluation points per algorithm. The winning algorithm was further analyzed through model-agnostic feature selection, with selection frequency tracked across folds and repetitions to identify stable predictors.

A final model instance was trained on all available data using optimal hyperparameters determined via 5-fold cross-validation, and its decision-making process was interpreted using SHAP (Shapley Additive Explanations) values.

1. Introduction

Coronary heart disease remains the leading cause of mortality worldwide, accounting for an estimated 19.8 million deaths in 2022 alone, representing approximately 32% of all global deaths. [1] Early and accurate identification of patients at risk is critical, as timely intervention can substantially reduce adverse outcomes including myocardial infarction and cardiac death. Traditional diagnostic pathways rely on a combination of clinical assessments, electrocardiographic findings, and invasive procedures such as coronary angiography, which carry inherent risks and costs. The application of machine learning to structured clinical data offers a promising, non-invasive complement to these methods, with the potential to support clinicians in triaging and risk-stratifying patients efficiently.

2. Materials & Methods

2.1. Dataset Overview

As previously mentioned, the dataset used for this project was UCI Heart Dataset which was forked and then analyzed in order to gain a better understanding of it. Below the results of the exploratory analysis.

DESCRIPTIVE STATISTICS OF ALL FEATURES

STATISTIC	<u>age</u>	<u>Sex</u>	<u>cp</u>	<u>trestbps</u>	<u>chol</u>	<u>lbs</u>	<u>restecg</u>	<u>thalach</u>	<u>exang</u>	<u>oldpeak</u>
COUNT	242	242	242	242	242	242	242	242	242	242
MEAN	54.55	0.68	3.15	130.96	249.84	0.14	0.98	149.96	0.33	1.00
STD	9.00	0.47	0.98	17.62	52.85	0.35	1.00	22.69	0.47	1.12
MIN	29.00	0.00	1.00	94.00	126.00	0.00	0.00	71.00	0.00	0.00
25%	48.00	0.00	2.25	120.00	212.00	0.00	0.00	134.50	0.00	0.00
50%	56.00	1.00	3.00	130.00	244.50	0.00	0.00	153.50	0.00	0.80
75%	61.00	1.00	4.00	140.00	277.75	0.00	2.00	166.00	1.00	1.60
MAX	77.00	1.00	4.00	200.00	564.00	1.00	2.00	202.00	1.00	6.20

STATISTIC	SLOPE	CA	THAL	NUM
COUNT	242.00	241.00	241.00	242.00
MEAN	1.59	0.61	4.71	0.46
STD	0.61	0.88	1.95	0.50
MIN	1.00	0.00	3.00	0.00
25%	1.00	0.00	3.00	0.00
50%	2.00	0.00	3.00	0.00
75%	2.00	1.00	7.00	1.00
MAX	3.00	3.00	7.00	1.00

Concerning the missing values of the dataset, only two were found, one in “ca” column and the other in “thal”.

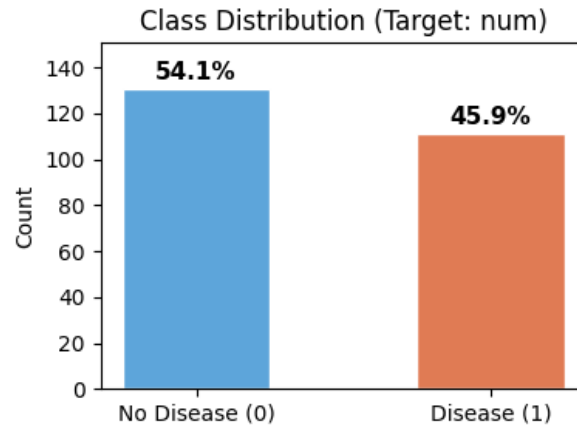
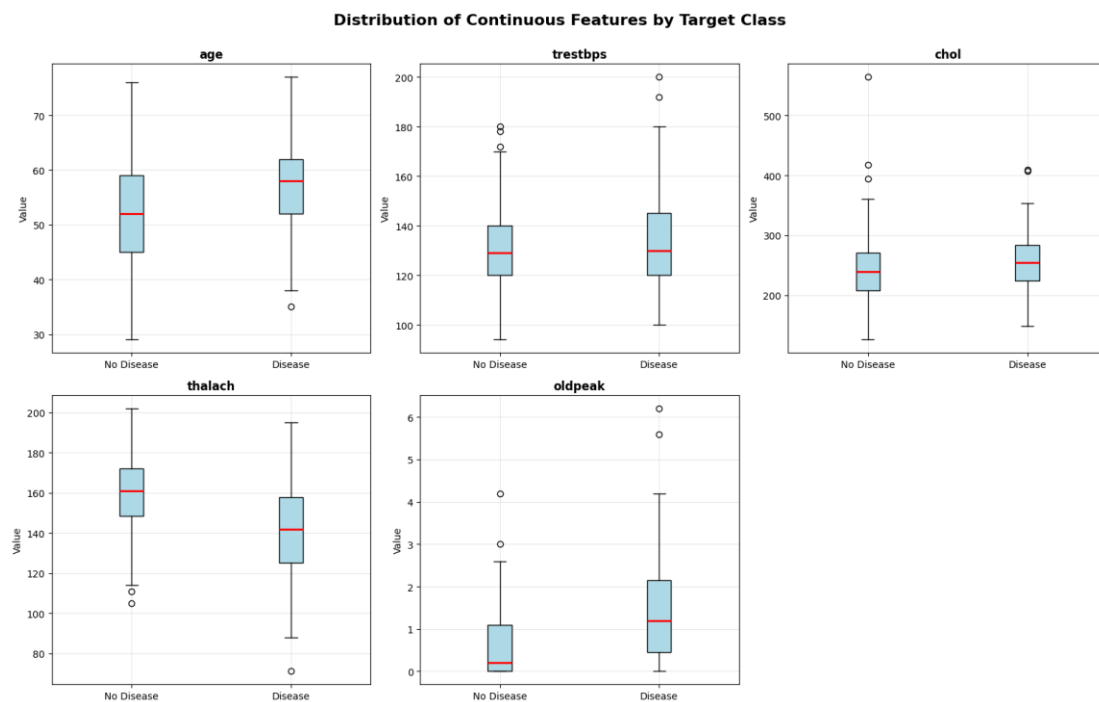


Figure 2-1 Class distribution of target column "num"

Reviewing the above plot, we can see a balanced set of the target column num between the two classes.

Furthermore, an outlier search was done on the continuous features and several were identified using the IQR method. In detail:

- Feature 'age': 0 outliers detected (0.00%)
- Feature 'trestbps': 6 outliers detected (2.48%)
- Feature 'chol': 5 outliers detected (2.07%)
- Feature 'thalach': 1 outliers detected (0.41%)
- Feature 'oldpeak': 4 outliers detected (1.65%)



In the above plot there can also be observed some of the outliers.

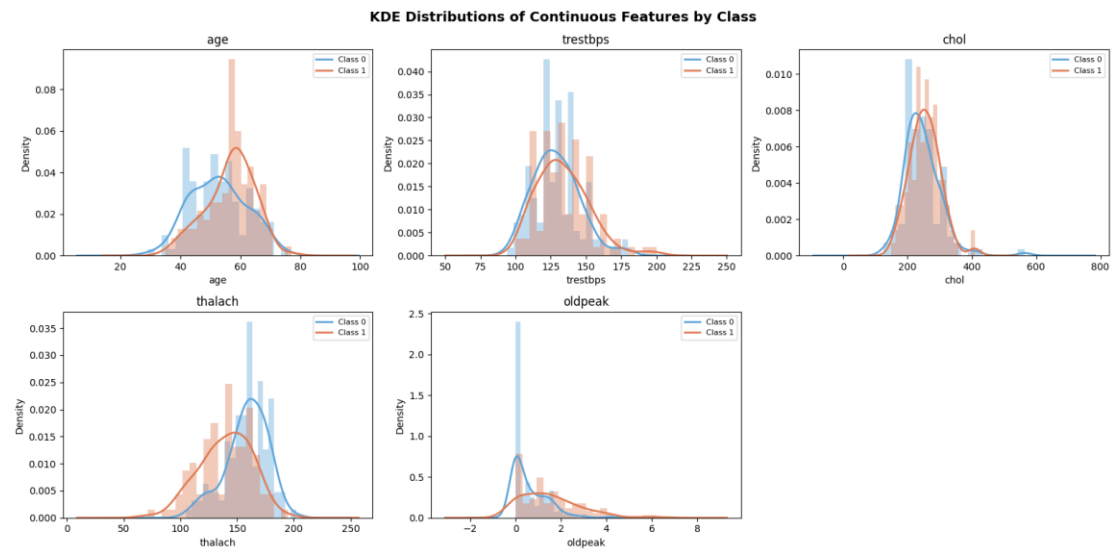


Figure 2-2 KDE Distributions of Continuous Features by class

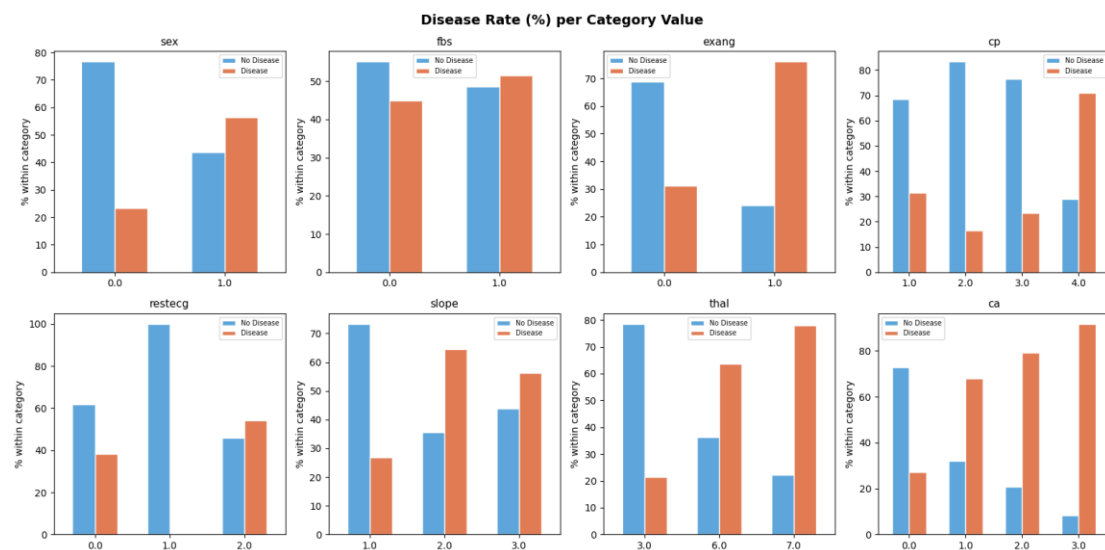


Figure 2-3 Disease Rate (%) per Category Value

Based on the two plots, several class imbalances can be observed such as difference between male and female patients or at the restecg variable where there are no patients with the disease having a class 1 ECG result.

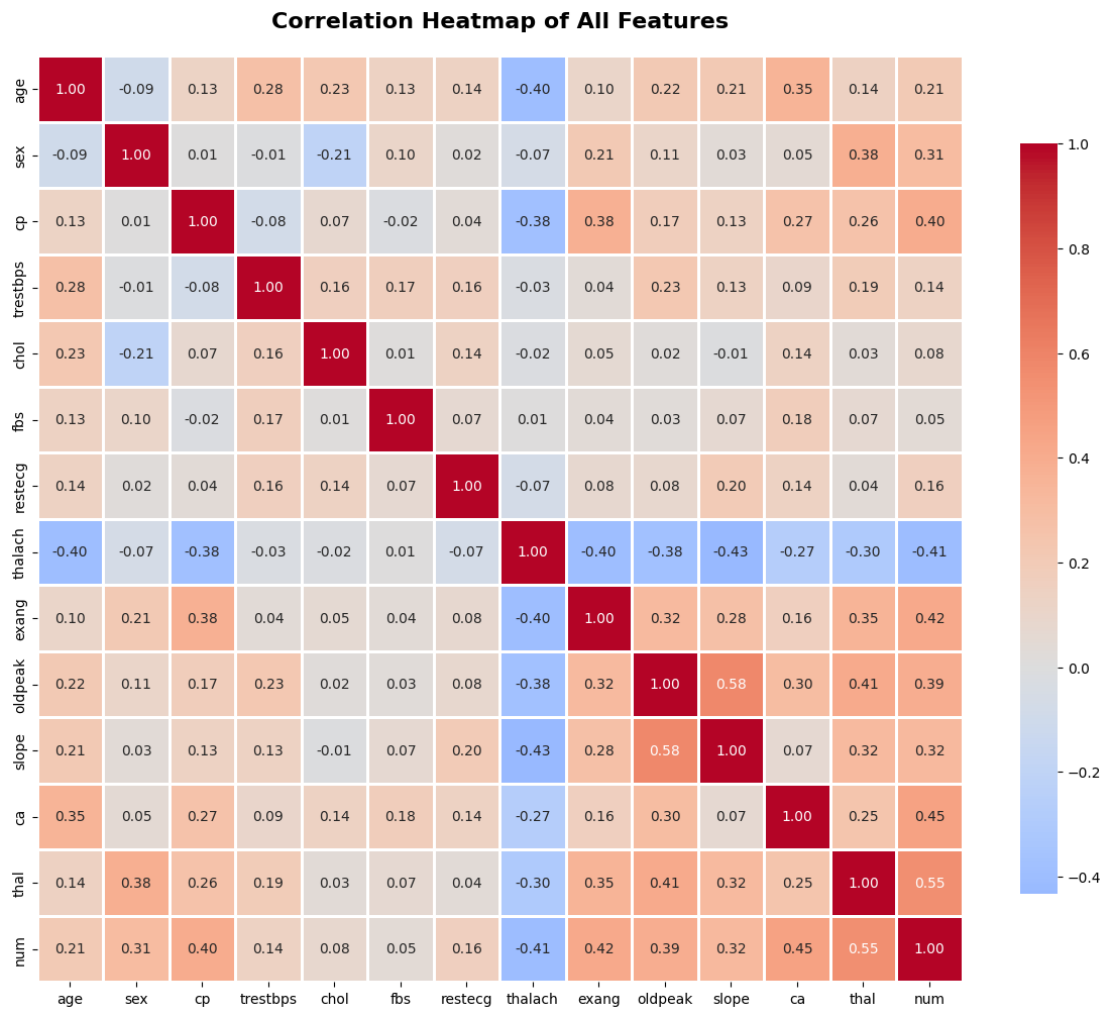


Figure 2-4 Correlation Heatmap of all features

Based on the pearson correlation matrix there are several strong correlations between our data. First of all, between the other features and the target column, “thal” (0.55) and “ca” (0.45) followed by “exang” (0.42) show a good positive correlation while on the other hand, “thalach” is the worst (-0.41) not only concerning the target variable but also most of the others. Some notable correlations are between “slope” and “oldpeak” (0.58) and “thalach” with “age” (-0.40) showing a really bad correlation. “Chol” and “fbs” seem to show the most neutral correlation.

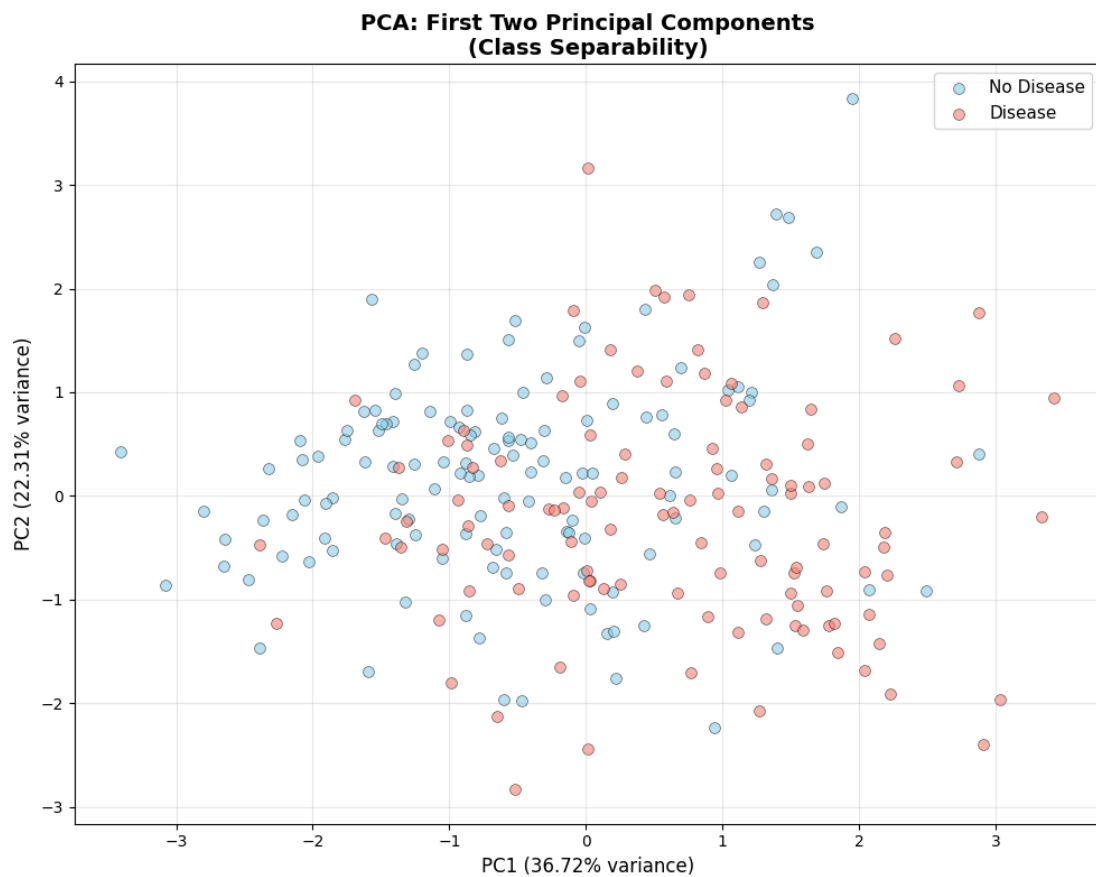


Figure 2-5 PCA plot of continuous features colored by target

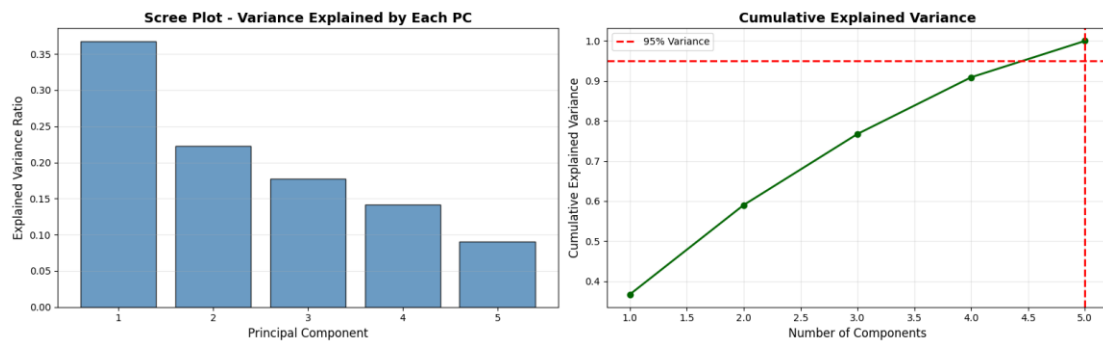
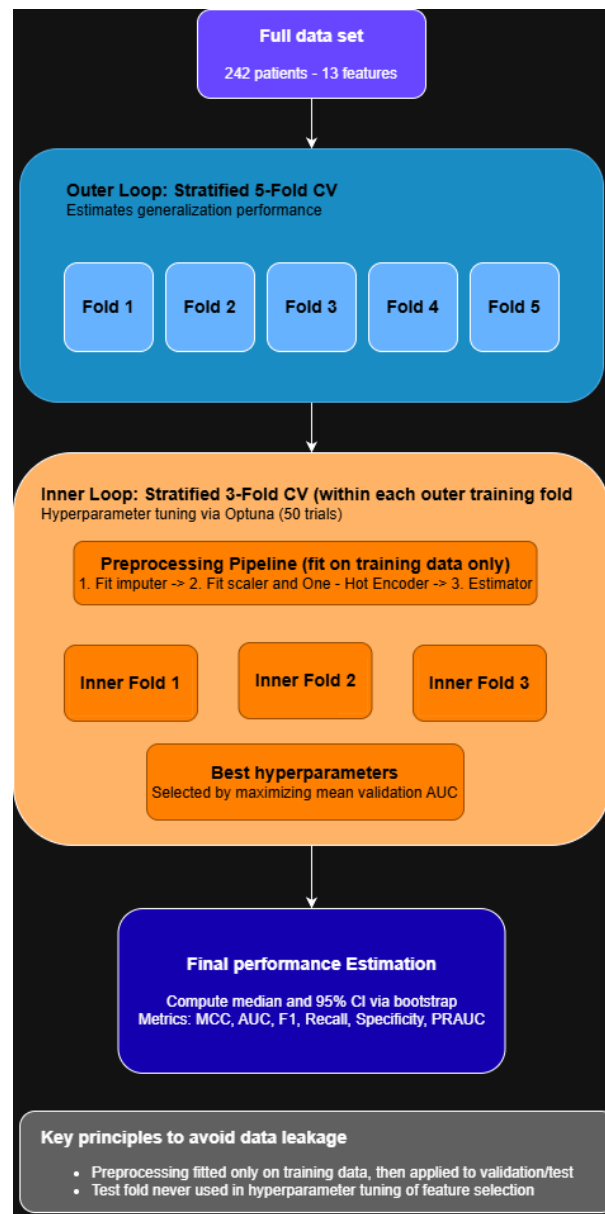


Figure 2-6 PCA variance

After the PCA reduction performance, a variance ratio of [0.37 0.22] was retrieved with the above plot showing an overlap between the two target classes and confirming that the numeric features likely carry meaningful signal.

2.2. Rncv Diagram

Before moving forward with the implementation of the pipeline, the following diagram was designed via app.diagrams.net in order to visualize the stages of it, avoiding data leakage and unnecessary steps.



2.3. The Repeated Nested Cross-Validation Class

A `RepeatedNestedCV` class was implemented in order to estimate the performance of each classification algorithm. The class takes as input the feature data, the target variable, a dictionary of the estimators and a dictionary containing the hyperparameters spaces of each estimator. Class balance was inspected prior to training. Due to moderate class imbalance, Matthews Correlation Coefficient (MCC) was adopted as the primary evaluation metric, as it is

invariant to class distribution and penalizes both false-positive and false-negative errors simultaneously.

For the data handling, a preprocessing pipeline was developed to avoid data leakage. It begins by imputing the numeric features with the median and categorical data with the most frequent, followed by a StandardScaler for the numeric data and a One-Hot Encoder for the categorical which is necessary for the tree algorithms. For each outer training fold, Optuna instantiated a fresh study using the TPE (Tree-structured Parzen Estimator) sampler seeded from the outer fold's random state. The study maximized mean ROC-AUC over K = 3 inner folds across 50 trials. The best parameter dictionary was then used to instantiate the estimator for outer-fold evaluation.

2.4. Hyperparameter spaces

Each model's search space was defined as an Optuna suggest function returning a parameter dictionary. Spaces were kept tractable (no redundant parameters) and model-appropriate.

MODEL	TUNED PARAMETERS	RANGE / CHOICES
LR	C	log-uniform [10^{-3} , 10]
LR	l1_ratio	uniform [0, 1]
GNB	var_smoothing	log-uniform [10^{-10} , 10^{-1}]
LDA	solver	fixed: lsqr
RF	n_estimators	{100, 200, 300, 400, 500}
	max_depth	int [3, 15]
	min_samples_split	int [2, 20]
	max_features	{sqrt, log2}
LIGHTGBM	n_estimators	{100, 200, 300, 400, 500}
	max_depth	Int [3,10]
	num_leaves	Int [16,64]
	learning_rate	Log-uniform [10^{-3} , 0.3]
	subsample	Float [0.6,1.0]
	colsample_bytree	Float [0.6,1.0]
XGBOOST	n_estimators	{100, 200, 300, 400, 500}
	max_depth	Int [3,10]
	learning_rate	Log-uniform [10^{-3} , 0.3]
	subsample	Float [0.6,1.0]
	colsample_bytree	Float [0.6,1.0]
	scale_pos_weight	fixed: n_neg / n_pos (class imbalance)
CATBOOST	Iterations	{100, 200, 300, 400, 500}
	Depth	Int [3,8]
	learning_rate	Log-uniform [10^{-3} , 0.3]
	l2_leaf_reg	Log-uniform [10^{-2} , 10]
	auto_class_weights	Balanced

Tree-based ensembles used class_weight = 'balanced' (RF, LightGBM) or auto_class_weights = 'Balanced' (CatBoost) to compensate for class imbalance. XGBoost instead used scale_pos_weight, computed as the ratio of negative to positive instances in the full dataset.

2.5. Feature selection

With 13 input features some may be redundant, noisy, or weakly discriminative. Reducing the feature set can improve model interpretability, reduce overfitting risk, and potentially improve generalization.

In this case, feature importance was estimated via permutation importance method, a model-agnostic technique that measures the decrease in ROC-AUC when a feature's values are randomly shuffled. This approach:

- Operates on the full fitted pipeline (including preprocessor), so importance scores correspond to original input features, not encoded/scaled representations.
- Is evaluated exclusively on the outer training fold, preventing any test-set contamination.
- Averages over 5 permutation repeats per fold to reduce variance in importance estimates.

The procedure is encapsulated in the `_select_features()` function within `nested_cv_feature_selection.py`

For the integration into the `rnCV` loop, the feature selection was embedded inside the outer loop of the `RepeatedNestedCV` class, before hyperparameter tuning. Crucially, both the inner-loop tuning and the final train/evaluate step operate on the same selected feature subset per fold.

2.6. SHAP Analysis

To finalize the model interpretation, SHAP values were used to evaluate the contribution of each feature in the model's predictions. Further results will be discussed in the following chapter.

3. Results

3.1. Baseline comparisons

After the first run having default parameters, the following results were retrieved showing that the LogisticRegression estimator had the best results (median values were calculated) showing the highest MCC followed by LDA with a small difference. The rest showed similar results with the worst model (at this point at least) the GNB.

	MCC	AUC	BA	F1	RECALL	SPECIFICITY	PRECISION	PRAUC
ESTIMATOR								
LR	0.6643	0.9015	0.8280	0.8090	0.8182	0.8846	0.8333	0.8986
GNB	0.5314	0.8833	0.7616	0.7524	0.8636	0.6603	0.6897	0.8575
LDA	0.6302	0.8902	0.8094	0.7907	0.7727	0.8846	0.8333	0.8928
RF	0.5882	0.8880	0.7885	0.7727	0.7727	0.8462	0.7958	0.8924
LIGHTGBM	0.5446	0.8525	0.7722	0.7500	0.7332	0.8077	0.7550	0.8452
XGBOOST	0.5854	0.8741	0.7920	0.7780	0.7727	0.8148	0.7860	0.8737
CATBOOST	0.5878	0.8794	0.7920	0.7710	0.7727	0.8462	0.8000	0.8811

3.2. Tuned models comparison

Once again, the LogisticRegression estimator was on top with GNB showing a great improvement with an increase of 0.1 on the MCC even outperforming the LDA as the previously second-best model. LogisticRegression had a slight decrease on several metrics such as Recall, indicating that the model is a bit less sensitive in identifying true positive cases. XGBoost was another model that showed a slight improvement surpassing the RF and reaching LDA.

	MCC	AUC	BA	F1	RECALL	SPECIFICITY	PRECISION	PRAUC
ESTIMATOR								
LR	0.6497	0.8990	0.8180	0.7958	0.7777	0.8846	0.8461	0.9016
GNB	0.6497	0.9007	0.8191	0.8000	0.7727	0.8846	0.8417	0.8965
LDA	0.6302	0.8902	0.8094	0.7907	0.7727	0.8846	0.8333	0.8928
RF	0.5859	0.8825	0.7860	0.7755	0.7391	0.8462	0.8000	0.8938
LIGHTGBM	0.5854	0.8756	0.7920	0.7710	0.7559	0.8305	0.8000	0.8843
XGBOOST	0.6238	0.8813	0.8007	0.7710	0.7332	0.8519	0.8333	0.8838
CATBOOST	0.6136	0.8829	0.7988	0.7815	0.7727	0.8682	0.8248	0.8891

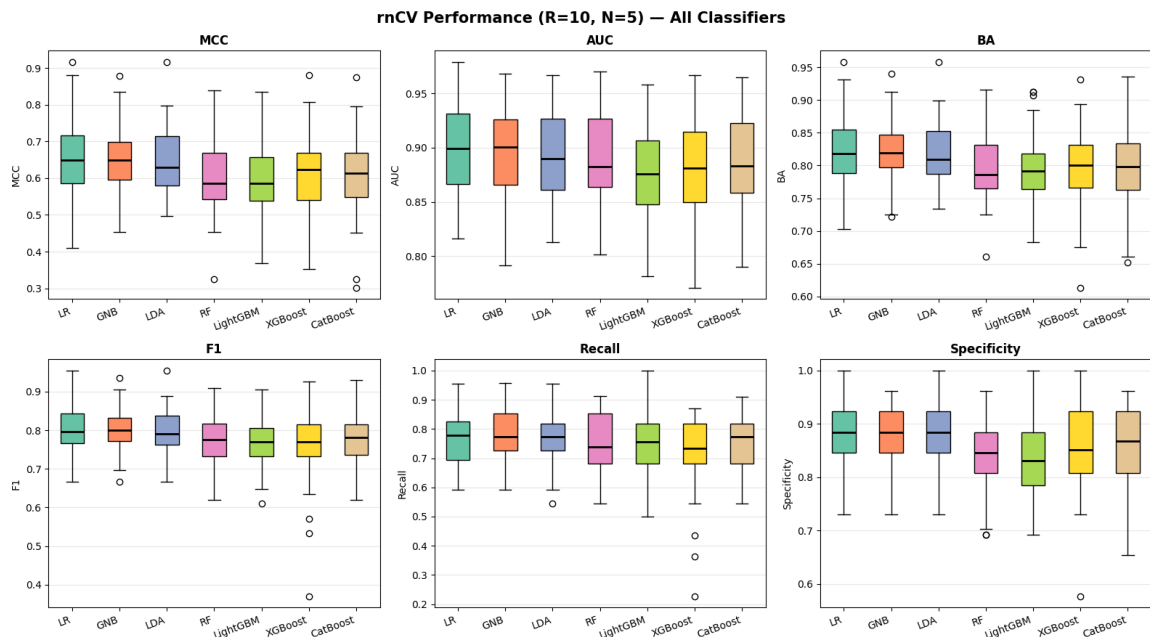


Figure 3-1 rnCV Boxplots

3.3. Feature selection on winner model

After selecting the LogisticRegression model as the best one among all, we re run the pipeline using a feature selection method, permutation importance. For a selected number of features (K=10) the algorithm showed a slight improvement concerning MCC (from 0.64 to 0.66). All the other metrics showed no drastic improvements.

	MCC	AUC	BA	F1	RECALL	SPECIFICITY	PRECISION	PRAUC
ESTIMATOR								
LR	0.6698	0.8996	0.8287	0.8095	0.7727	0.8846	0.85	0.8974

	FULL (13 FEATS)	SELECTED (10 FEATS)	DELTA
METRIC			
MCC	0.6692 [0.6044, 0.6800]	0.6698 [0.6197, 0.7089]	0.0006
AUC	0.9038 [0.8795, 0.9141]	0.8996 [0.8820, 0.9100]	-0.0042
BA	0.8269 [0.8022, 0.8357]	0.8287 [0.8064, 0.8479]	0.0018
F1	0.8048 [0.7860, 0.8205]	0.8095 [0.7860, 0.8293]	0.0047
RECALL	0.7727 [0.7273, 0.8182]	0.7727 [0.7391, 0.8182]	0.0000
SPECIFICITY	0.8846 [0.8462, 0.9060]	0.8846 [0.8462, 0.9060]	0.0000

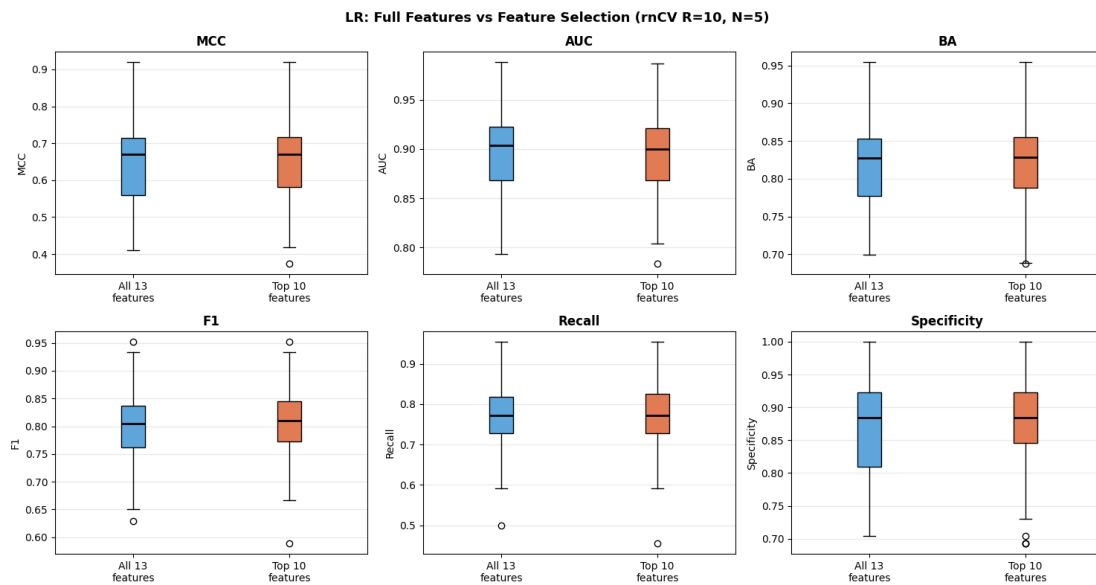


Figure 3-2 Feature selection comparison

3.4. Feature selection frequency

For the table below five features appear to be the most stable and predictive ones. Among the top ones we should highlight that 4/5 are categorical data, indicating that they are a strong factor for models performance.

	SELECTED COUNT	SELECTED %
SEX	50	100.0
CP	50	100.0
SLOPE	50	100.0
THAL	50	100.0
CA	50	100.0
THALACH	49	98.0
TRESTBPS	45	90.0
RESTECG	35	70.0
CHOL	35	70.0
OLDPEAK	32	64.0
EXANG	30	60.0
AGE	13	26.0
FBS	11	22.0

3.5. Final model interpretation

After identifying the winner algorithm using rnCV, we perform a final 5-fold cross-validation now using the whole dataset to determine the optimal set of hyperparameters for the final model instance to deploy.

In order to identify the impact and the contribution to the model's prediction we calculate SHAP values. Based on the following plots, as also seen before, "ca", "cp", "thal", "sex" and "slope" are the top 5 most important features. In this particular case due to the One-Hot Encoding the categorical data are divided into multiple columns.

The most impactful one is ca_0.0, more specifically the ordinal factor ca which is the number of major vessels (0–3) colored by fluoroscopy. It should be noted though that not every category of this feature has the same impact on the model. Ca_1.0 – Ca_3.0 have extremely low to no importance. This can be observed also on different "slope" and "thal" categories.

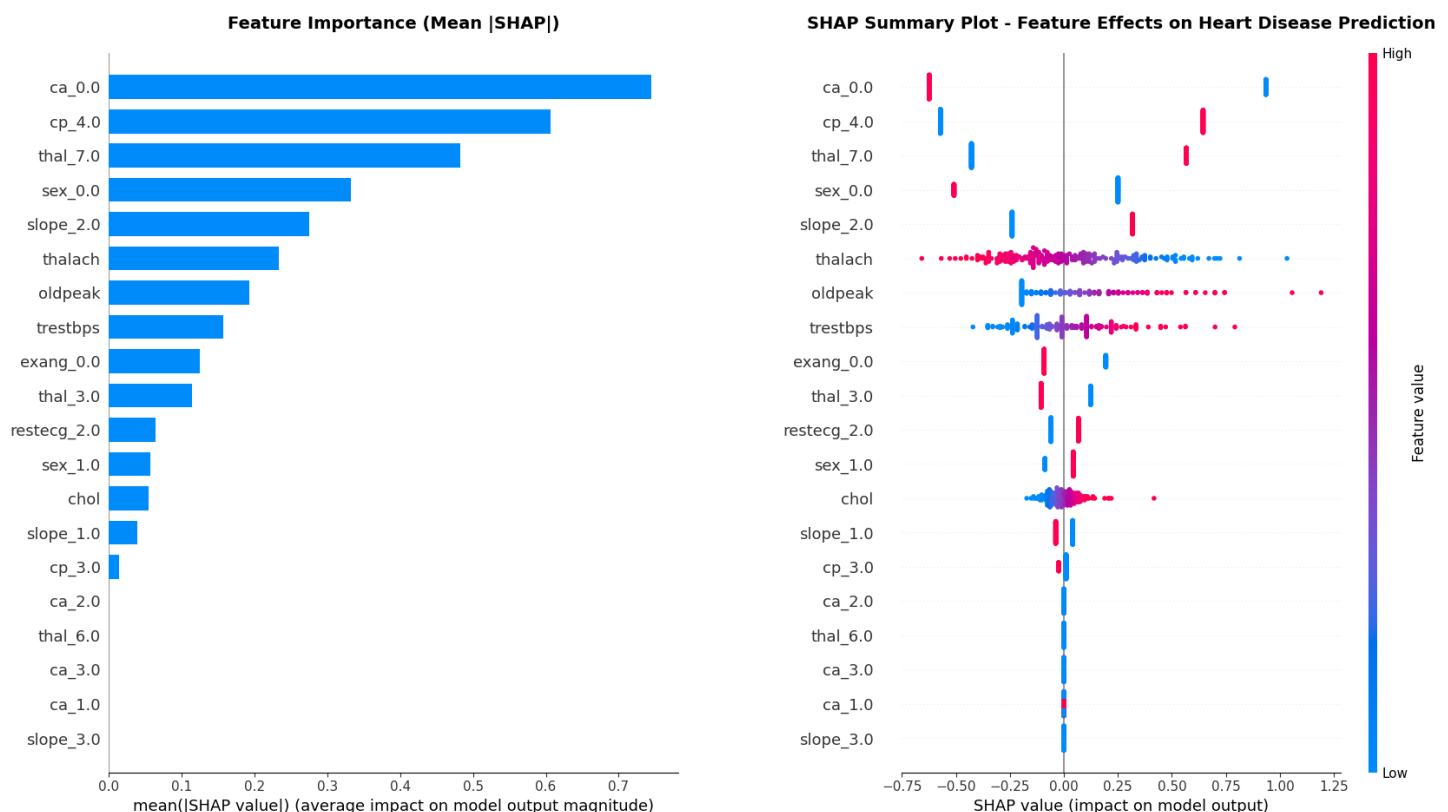


Figure 3-3 Summary plots for feature importance after calculation of SHAP Values

4. Conclusions

The project successfully established a robust machine learning pipeline for the binary classification of coronary artery disease using the UCI Heart Dataset. Through this whole process, several insights were uncovered. First of all, Logistic Regression with Elastic Net

regularization emerged as the most effective classifier, consistently achieving the highest Matthews Correlation Coefficient (MCC) in baseline tests and maintaining strong performance after hyperparameter tuning. Moving forward to the feature importance analysis, through both permutation importance and SHAP values a stable core of predictors was identified lead by the presence of 1-3 major blood vessels colored by fluoroscopy. Notably, categorical data proved to be the most influential factors in model performance.

While the pipeline was designed with rigorous methods to prevent data leakage and provide unbiased estimates, several limitations remain. The analysis was conducted on a relatively small subset (242 patient records). While sufficient for demonstrating the methodology, larger and more diverse datasets are required to ensure the model generalizes well across different populations and clinical settings. Also, some class imbalances were noted in categorical variables, such as “restecg”, where no diseased patients had a class 1 ECG result. This lack of representation for certain sub-categories can limit the model's ability to learn specific risk patterns.

5. Suggested further research

To build upon the current findings there are some areas that can be further researched. A Deep Learning comparison would definitely determine if neural networks can outperform traditional classifiers when given the same cross-validation treatment.

Finally, an investigation on advanced ensemble techniques such as stacking the linear models (LR, LDA) with non-linear tree-based models (XGBoost, CatBoost) would appear useful to see if combined architectures can further boost sensitivity without sacrificing specificity.

6. References

1. World Health Organization, “Cardiovascular disease (CVDs)”, 2023

7. AI usage

LLMs were used in the process of writing the code for purposes of debugging, leakage identification and general guidance.